

Meta-Learning-Driven Joint Power Allocation and Phase-Shift Design for RIS-Assisted NOMA-ISAC Systems

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Abstract—This letter investigates a reconfigurable intelligent surface (RIS)-assisted multiple-input single-output (MISO) integrated sensing and communication (ISAC) system, where a base station (BS) simultaneously serves a near and a far user via non-orthogonal multiple access (NOMA) while illuminating a sensing target. We maximize a weighted communication-and-sensing throughput by jointly optimizing the NOMA power-allocation coefficients and the RIS phase shifts, subject to an aggregate communication QoS floor, individual near/far-user QoS floors, and a sensing QoS constraint. To tackle the resulting non-convex problem, a model-agnostic meta-learning (MAML)-based deep-learning algorithm is proposed. A lightweight network predicts the low-dimensional power split, while an inner optimizer adapts the high-dimensional RIS phase vector for each channel realization. A first-order meta-update then trains the power map using the phase-adapted channels, avoiding the unstable higher-order derivatives observed in this high-SINR setting. Simulation results show stable convergence, a strong RIS-aperture gain, and consistent improvements against OMA, no-RIS, and partially optimized baselines.

Index Terms—Reconfigurable intelligent surface (RIS), integrated sensing and communication (ISAC), non-orthogonal multiple access (NOMA), deep learning, meta-learning (MAML), MISO.

I. INTRODUCTION

INTEGRATED sensing and communication (ISAC) is a key enabler for next-generation wireless systems because sensing and data transmission can share spectrum, antennas, and radio-frequency hardware [1]. Power-domain non-orthogonal multiple access (NOMA) further allows multiple users to occupy the same time-frequency resource by superposition coding and successive interference cancellation (SIC), improving spectral efficiency when users experience heterogeneous channel strengths [14], [12]. Reconfigurable intelligent surfaces (RISs) complement both ideas by controlling the propagation environment through programmable passive elements, which is particularly useful when the direct BS–user paths are blocked or severely attenuated [13], [3].

Recent RIS-assisted ISAC studies have mainly relied on iterative optimization. Hybrid active/passive beamforming was considered in [4], while [5] used block-coordinate optimization to improve sensing accuracy and [6] investigated RIS-assisted rate-splitting ISAC. For NOMA-assisted ISAC, [12]

established the benefits of sharing resources between sensing and NOMA communication. Learning-based RIS design is attractive because solving a non-convex phase-and-resource-allocation problem from scratch for every channel realization can be costly. In this direction, Zou *et al.* [7] applied machine learning and meta-learning to RIS-assisted NOMA resource optimization, while the general meta-learning literature shows how experience across related tasks can be used to obtain fast adaptation on a new task [9], [15].

The remaining challenge in the considered RIS-NOMA-ISAC setting is that the power split and RIS phases are tightly coupled: the three power coefficients are low dimensional, whereas the RIS phase vector grows with N and must react to each channel realization. This motivates a bilevel design in which a compact network learns the reusable power-allocation map and a short channel-specific inner loop adapts the RIS phases.

The contributions of this letter are threefold. First, we formulate joint power-allocation and RIS-phase optimization for MISO NOMA-ISAC with a communication-cluster QoS constraint, individual near/far-user QoS constraints, and a sensing QoS constraint. The individual constraints prevent a strong user from masking a weak user behind an acceptable NOMA sum rate. Second, we develop a first-order MAML-based bilevel solver in which the power-allocation network is trained across channel realizations after RIS-phase adaptation; the formulation therefore combines reusable learning with a small fixed online optimization budget. Third, we characterize the convergence limitation and online complexity, and evaluate the design against RIS/no-RIS, NOMA/OMA, and partial-optimization baselines using the same channel model and held-out test procedure.

II. SYSTEM MODEL AND PROBLEM FORMULATION

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

As illustrated in Fig. 1, an M -antenna ISAC BS serves a single-antenna near user n and far user f while illuminating a point target through an N -element RIS. The coefficient of RIS element i is written as

$$v_i = \rho_i e^{j\phi_i}, \quad 0 \leq \rho_i \leq 1, \quad \phi_i \in [0, 2\pi), \quad (1)$$

and $\Phi = \text{diag}(v_1, \dots, v_N)$. The formulation keeps ρ_i symbolic; the ideal passive-RIS value used in the simulations is reported only in Table I.

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System-model artwork placeholder. Upload `figs/fig_system_model.jpg` or the EPS version supplied with the revision.

Fig. 1. RIS-assisted MISO NOMA-ISAC system. The BS–RIS forward and reciprocal channels are denoted by $\mathbf{H}_{BR}$ and $\mathbf{H}_{RB}$, the RIS–user channels by $\mathbf{h}_{RD_n}$ and $\mathbf{h}_{RD_f}$, and the sensing links by $\mathbf{h}_{RT}$ and $\mathbf{h}_{TR}$.

1) *Channel Model*: Let $\mathbf{H}_{BR} \in \mathbb{C}^{N \times M}$ denote the BS–RIS channel and let $\mathbf{H}_{RB}$ denote the reciprocal RIS–BS channel. For user $k \in \{n, f\}$, $\mathbf{h}_{RD_k} \in \mathbb{C}^N$ denotes the RIS–user channel. The effective user channel is

$$\mathbf{h}_k = \mathbf{H}_{RB} \Phi \mathbf{h}_{RD_k} \in \mathbb{C}^M. \quad (2)$$

The direct BS–user paths are blocked in the RIS-assisted data set, so the cascaded term dominates. For the sensing path, $\mathbf{h}_{RT}$ and $\mathbf{h}_{TR}$ denote the RIS–target and target–RIS links, respectively. A rank-one round-trip target response is modeled as

$$\mathbf{G}_T = \sqrt{\beta_T} (\mathbf{H}_{RB} \Phi \mathbf{h}_{TR}) (\mathbf{h}_{RT}^H \Phi \mathbf{H}_{BR}), \quad (3)$$

where β_T is the target/path-gain coefficient. Numerical values for β_T , path loss, and fading are deliberately kept out of the model equations and are listed in Section IV.

2) *NOMA Communication and Sensing Model*: The transmitted signal is

$$\mathbf{x} = \left(\sqrt{a_n P} s_n + \sqrt{a_f P} s_f \right) \mathbf{w}_c + \sqrt{a_T P} \mathbf{w}_T s_T, \quad (4)$$

where $\mathbf{a} = [a_n, a_f, a_T]$ is the near-user, far-user, and sensing power split, $a_n + a_f + a_T = 1$, and all entries are nonnegative. The unit-norm common communication beam is parameterized without hard-coded numerical weights as

$$\mathbf{w}_c = \frac{\mu_f \mathbf{h}_f + \mu_n \mathbf{h}_n}{\|\mu_f \mathbf{h}_f + \mu_n \mathbf{h}_n\|}, \quad \mu_f, \mu_n \geq 0, \quad \mu_f + \mu_n = 1, \quad (5)$$

where μ_f and μ_n control the far/near steering emphasis. Both NOMA users share $\mathbf{w}_c$ and are separated through SIC [14], [7].

For sensing, define $\mathbf{H}_u = [\mathbf{h}_n, \mathbf{h}_f]$ and the leakage-aware matrix

$$\mathbf{A}_T = \mathbf{G}_T^H \mathbf{G}_T - \lambda_L \mathbf{H}_u \mathbf{H}_u^H. \quad (6)$$

The sensing beam is the normalized dominant eigenvector, $\mathbf{w}_T = \mathbf{v}_{\max}(\mathbf{A}_T) / \|\mathbf{v}_{\max}(\mathbf{A}_T)\|$. Thus λ_L is a tunable leakage-control parameter rather than a numerical constant embedded in the system model [4], [5].

With $g_{kj} = |\mathbf{h}_k^H \mathbf{w}_j|^2$, the SINRs used by the implementation are

$$\gamma_f = \frac{a_f g_{ff} P}{a_n g_{ff} P + a_T g_{fT} P + \sigma^2}, \quad (7)$$

$$\gamma_n = \frac{a_n g_{nn} P}{a_T g_{nT} P + \sigma^2}, \quad (8)$$

$$\gamma_s = \frac{a_T |\mathbf{u}_T^H \mathbf{G}_T \mathbf{w}_T|^2 P}{\sigma^2}, \quad (9)$$

where σ^2 is the receiver-noise power and $\mathbf{u}_T$ is the sensing receive combiner. The rates are $R_k = \log_2(1 + \gamma_k)$, $k \in \{n, f, s\}$. The weighted ISAC objective is

$$R_{DL} = \omega_c (R_n + R_f) + \omega_s R_s, \quad \omega_c + \omega_s = 1. \quad (10)$$

The sum $R_n + R_f$ is the NOMA-cluster spectral efficiency, while ω_c and ω_s express the communication/sensing design priority. Their numerical settings are reported with the simulation parameters.

B. Problem Formulation

We jointly optimize the power split $\mathbf{a}$ and RIS phases $\Phi = [\phi_1, \dots, \phi_N]$. The communication sum rate is retained as a cluster-level QoS measure, while individual user constraints are added in the same spirit as the per-device QoS constraint in (9b) of [7]. The resulting problem is

$$(P1) \quad \max_{\mathbf{a}, \Phi} \quad R_{DL} \quad (11a)$$

$$\text{s.t.} \quad R_n + R_f \geq R_{th,c}, \quad (11b)$$

$$R_k \geq R_{th,k}, \quad k \in \{n, f\}, \quad (11c)$$

$$R_s \geq R_{th,s}, \quad (11d)$$

$$a_n + a_f + a_T = 1, \quad a_n, a_f, a_T \geq 0, \quad (11e)$$

$$|e^{j\phi_i}| = 1, \quad i = 1, \dots, N. \quad (11f)$$

The aggregate constraint (11b) is useful because the two NOMA users share one time-frequency resource block, so $R_n + R_f$ directly measures the communication service delivered by the cluster. It also matches the cluster-level QoS term used by the implemented training loss. On its own, however, a sum threshold is insufficient: a very large near-user rate could compensate for an unacceptable far-user rate. Constraint (11c) therefore adds the individual protection requested in the formulation, exactly analogous to writing a QoS condition for each device as in (9b) of [7]. Thus, the sum threshold controls aggregate communication quality while the individual thresholds prevent user starvation. The sensing requirement is enforced separately by (11d). The ideal-RIS assumption $\beta_i = 1$ is fixed rather than optimized, and the phase parameterization $e^{j\phi_i}$ enforces unit modulus automatically.

Problem (P1) remains non-convex because the power variables and RIS phases are coupled through the cascaded channels. We therefore enforce the QoS conditions through penalty terms during learning, while the softmax output enforces the power simplex exactly.

IV. PROPOSED MAML-BASED SOLUTION

A. Motivation for Meta-Learning

The optimization variables react on two different time and dimensional scales. The power vector has only three entries and exhibits reusable structure across related channel realizations, whereas the N RIS phases are strongly channel-specific. A conventional alternating optimizer would solve the coupled non-convex problem again for every realization, while a purely one-shot neural network must learn a direct map to a high-dimensional phase vector and can be sensitive to distribution shifts.

Meta-learning is well suited to this structure because it learns across a family of related tasks and is explicitly designed for fast adaptation on a new task [9], [15]. Here each channel realization is treated as an adaptation task: the outer learner captures the reusable power-allocation rule, and the inner loop spends a small fixed number of gradient steps adapting the RIS phase to the current channel. This learning-plus-adaptation structure follows the motivation of RIS-NOMA meta-learning in [7], but is specialized to the joint communication-and-sensing objective and its QoS penalties. The implemented first-order update further avoids storing the full higher-order derivative graph, reducing training memory and improving numerical robustness [10], [11].

B. Feature Map and Power-Allocation Network

For a given phase vector, the cascaded channels and the six effective gains $\{g_{nn}, g_{ff}, g_{sT}, g_{fn}, g_{fT}, g_{nT}\}$ are computed. They are converted to dB and concatenated with the communication/sensing QoS levels and the SNR budget, giving a nine-dimensional input feature vector $\mathbf{z}$. The power network contains three hidden layers with 256 neurons per layer, LayerNorm and LeakyReLU activations, followed by a three-neuron softmax head. Hence

$$\mathbf{a} = \text{softmax}(f_{\theta}(\mathbf{z})), \quad (12)$$

which enforces $a_n + a_f + a_T = 1$ and $a_k \geq 0$ by construction.

C. Joint Power-Allocation and Phase Adaptation

For a channel sample and the current phase vector $\phi^{(j)}$, the network first predicts

$$\mathbf{a}^{(j)} = \text{softmax}\left(f_{\theta}(\mathbf{z}(\phi^{(j)}))\right). \quad (13)$$

The implemented constrained loss is

$$\begin{aligned} \ell^{(j)} = & -R_{\text{DL}} + \lambda_c (R_{\text{th},c} - R_n - R_f)_+ + \lambda_s (R_{\text{th},s} - R_s)_+ \\ & + \lambda_n (R_{\text{th},n} - R_n)_+ + \lambda_f (R_{\text{th},f} - R_f)_+, \end{aligned} \quad (14)$$

where $(x)_+ = \max(x, 0)$. Equation (14) makes explicit why the aggregate and individual communication thresholds coexist: λ_c protects cluster throughput, whereas λ_n and λ_f penalize starvation of either user.

The two update equations below describe the inner phase adaptation and the outer meta-update. The index $j \in \{0, \dots, K-1\}$ denotes the inner step, K is the fixed number of RIS-adaptation steps, η_{ϕ} is the inner phase-learning rate, θ denotes the parameters of the power-allocation network, α is the

outer learning rate, and B is the mini-batch size. The penalty coefficients $\lambda_c, \lambda_s, \lambda_n, \lambda_f$ control the strength of the aggregate-communication, sensing, near-user, and far-user QoS penalties, respectively.

The RIS phase is represented by $\theta_i = e^{j\phi_i} = \cos \phi_i + j \sin \phi_i$, so unit modulus is satisfied for any real ϕ_i . For each channel, the phase vector is randomly initialized and adapted for K steps using Adam on (14),

$$\phi^{(j+1)} = \text{Adam}_{\phi}\left(\phi^{(j)}, \nabla_{\phi^{(j)}} \ell^{(j)}; \eta_{\phi}\right), \quad (15)$$

with inner learning rate η_{ϕ} . Since the effective gains change after every phase update, the feature vector and power split are recomputed at each inner step.

D. First-Order Meta-Update

After the K phase-adaptation steps, the final rates and loss are recomputed at $\phi^{(K)}$. The implementation uses a first-order MAML approximation: the adapted phase is treated as fixed during the outer network update rather than backpropagating second-order derivatives through the entire Adam trajectory. The outer update is therefore

$$\theta \leftarrow \theta - \alpha \nabla_{\theta} \frac{1}{B} \sum_{b=1}^B \ell_b(\theta, \phi_b^{(K)}), \quad (16)$$

where the gradient is evaluated at the phase-adapted state $\phi_b^{(K)}$. Thus, Eq. (15) changes the channel-specific RIS variable while Eq. (16) updates the reusable network parameters. The former uses η_{ϕ} and K ; the latter uses the mini-batch size B and outer rate α . This retains the key meta-learning idea—the power map is trained on channels after per-instance RIS adaptation—while avoiding the memory and numerical burden of the full higher-order graph. In the implemented fair, high-SINR configuration, the full second-order path was found to produce unstable/non-finite meta-gradients, which motivated the first-order variant.

E. Training and Prediction Procedure

Algorithm 1 summarizes the implementation. At inference, the same K -step phase adaptation is performed for an unseen channel and the final softmax output provides the power allocation. Thus prediction is not a pure one-shot neural-network call; it is a short learned-plus-optimization procedure that trades a small fixed online cost for substantially better phase adaptation.

F. Convergence Analysis

A global convergence guarantee is not claimed. The objective is non-convex, the unit-modulus RIS parameterization couples all rates through the cascaded channel, and the hinge penalties in (14) are non-smooth exactly at the QoS boundaries. Moreover, as discussed for the related RIS-NOMA MAML method in [7], standard smooth MAML convergence theorems do not directly cover an architecture in which the inner and outer loops optimize different variables. Our first-order approximation further discards the higher-order dependence of the adapted phase on θ . Consequently, convergence is assessed empirically: the inner Adam loop is run for a fixed number of steps and the outer training curve is checked for a stable loss/rate plateau across the reported hyperparameter settings.

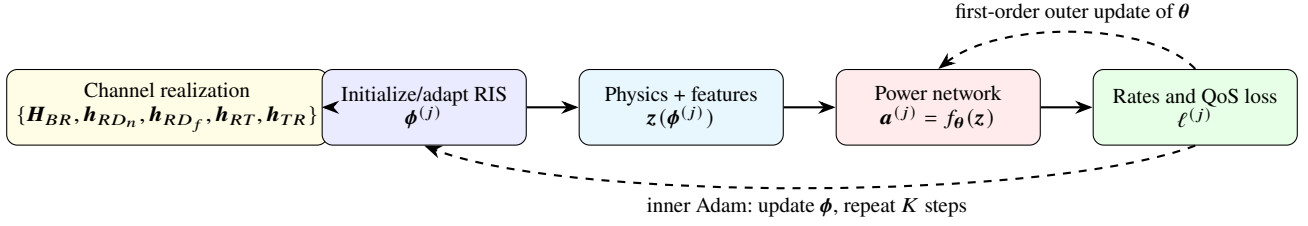


Fig. 2. First-order MAML-based joint power/phase design. The inner loop adapts the channel-specific RIS phase and recomputes the power split; after K inner steps, the outer update trains the reusable power-allocation network. The two-column layout separates the inner and outer update paths to avoid overlap.

Algorithm 1: First-order MAML-based power and RIS-phase optimization

Input: channel data; inner steps K ; inner Adam rate η_ϕ ; outer rate α ; penalties $\lambda_c, \lambda_s, \lambda_n, \lambda_f$.

Output: trained power-map parameters θ^* .

Initialize θ

while not converged do

 Sample a mini-batch of B channels

foreach channel b **do**

 Initialize $\phi_b^{(0)} \sim \mathcal{U}([0, 2\pi)^N)$

for $j = 0$ **to** $K - 1$ **do**

 Compute phase-dependent channels/gains and $z_b^{(j)}$

 Predict $a_b^{(j)}$ using (13)

 Evaluate R_n, R_f, R_s and (14)

 Update $\phi_b^{(j+1)}$ with inner Adam

end

 Recompute $a_b^{(K)}$ and final loss at the adapted phase

end

 Update θ with Adam using the first-order gradient in (16)

end

Select θ^* by validation score

TABLE I
SYSTEM AND CHANNEL PARAMETERS.

Parameter	Value
BS antennas M	4
RIS elements N	8 nominal; {8, 16, 32, 64} in N sweep
BS location	[0, 0, 0] m
RIS location	[8, 0, 0] m
Near-user location	[9.69, 2.48, 0] m
Far-user location	[4.00, 24.68, 0] m
Target location	[8.00, 6.00, 0] m
Reference gain PL_0	10^{-3} (-30 dB)
Used path-loss exponents	$\alpha_{BR} = 2.2$, $\alpha_{RU} = 2.8$, $\alpha_{RT} = 2.3$
Nakagami fading classes	user-facing: $m = 1$; infrastructure/sensing: $m = 2-3$
Target coefficient β_T	40 dB
Ideal RIS amplitude ρ_i	1
Bandwidth / temperature / NF	1 MHz / 290 K / 6 dB
Nominal generator power	40 dBm
Power sweep	5, 10, 15, 20, 25 dBm

TABLE II
LEARNING AND OPTIMIZATION PARAMETERS.

Parameter	Value
Data partition	90% train / 10% validation; independent test set
Power network	9-256-256-256-3, softmax output
Outer optimizer	Adam
NOMA-RIS training	batch 64, lr 10^{-4} , 40 epochs
Inner phase adaptation	$K = 10$, Adam, $\eta_\phi = 0.1$
Beam weights (μ_f, μ_n)	(0.7, 0.3) in simulations
Leakage parameter λ_L	0.5 in simulations
Objective weights (ω_c, ω_s)	(0.7, 0.3)
Cluster/sensing QoS	$R_{th,c} = 21.7$, $R_{th,s} = 2.9$ bit/s/Hz
Individual user floors	$R_{th,n} = R_{th,f} = 1$ bit/s/Hz
Penalty weights $(\lambda_c, \lambda_s, \lambda_n, \lambda_f)$	(2, 50, 15, 20)

G. Computational Complexity

Let P_θ denote the power-network parameter count. With nine inputs, three 256-unit hidden layers, LayerNorm, and a three-output head, the implemented network contains approximately 1.36×10^5 trainable parameters. Computing the user cascades costs $\mathcal{O}(MN)$, while forming the sensing matrix and its $M \times M$ leakage-aware eigenvector adds approximately $\mathcal{O}(MN + M^3)$. Therefore, one inner prediction/adaptation step costs

$$\mathcal{O}(P_\theta + MN + M^3), \quad (17)$$

and online prediction for one channel realization costs

$$\mathcal{O}(K(P_\theta + MN + M^3)), \quad (18)$$

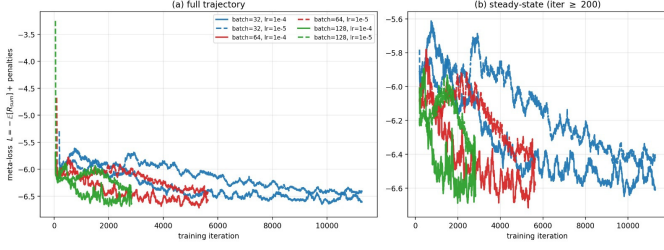
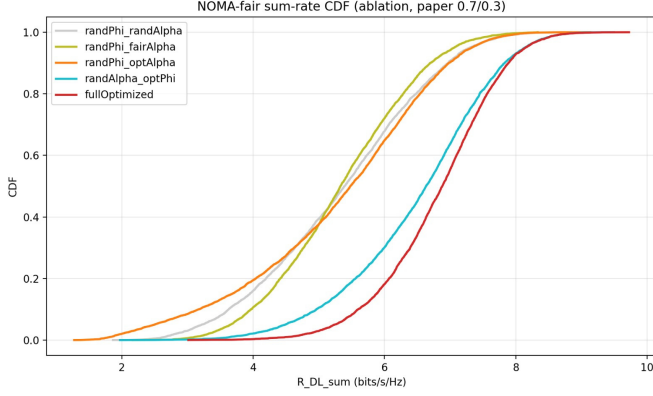
plus one final forward evaluation of the same order. For a batch of B samples, an offline training iteration scales as $\mathcal{O}(BK(P_\theta + MN + M^3))$. Because the implemented outer update is first-order, it does not store a full second-order graph through all K inner steps, reducing memory and derivative cost relative

to full MAML. This is the relevant complexity for *prediction*: one compact network plus a fixed, short RIS adaptation, rather than solving a fresh long alternating-optimization problem from scratch for every channel.

V. NUMERICAL RESULTS AND ANALYSIS

The MATLAB generator specifies pairwise link lengths rather than absolute coordinates. To report the geometry in the requested $[x, y, z]$ form, Table I uses one equivalent coordinate realization that reproduces the simulated BS-RIS, RIS-user, BS-user, and RIS-target separations; rotations or reflections of this embedding are physically equivalent because only the link lengths enter the path-loss model. Direct BS-user links are disabled in the RIS-assisted data set, so their path-loss exponents are not listed.

The aggregate NOMA threshold in Tables I and II comes from the fair NOMA dataset, where it was chosen to keep the cluster QoS active; the individual near/far floors are the defaults

Fig 4 — NOMA MAML training loss vs iterations (batch x lr sweep, $M=4$ fair, $N=8$)

 Fig. 3. MAML meta-loss versus training iteration over a batch/learning-rate sweep ($M=4$, $N=8$). (a) Full trajectory; (b) steady state (iter ≥ 200).

 Fig. 4. CDF of the weighted sum-rate R_{DL} over the test set for the five ablation configurations.

used by the training scripts. The OMA fair dataset uses its own calibrated aggregate/sensing thresholds ($R_{th,c} = 12$ and $R_{th,s} = 2$ bit/s/Hz). The service-rate outage thresholds stored by the MATLAB generators are diagnostic quantities and are not the same as the optimization penalties in (14).

The implementation records per-iteration loss and weighted rate for convergence monitoring, while model selection uses a held-out validation score. The present analysis therefore relies on the fixed-step inner optimizer, validation-based checkpointing, and the convergence limitations discussed in Section III-E rather than asserting a closed-form global guarantee.

Fig. 3 shows the MAML meta-loss (14) over training iterations for several batch/learning-rate settings. The loss decreases smoothly and the weighted sum-rate climbs steadily, with no epoch-one collapse, indicating stable first-order meta-optimization; the steady-state panel confirms continued, generalizing improvement.

Fig. 4 plots the empirical CDF of the weighted sum-rate R_{DL} over the test set for the ablation. Joint optimization of phase and power (fullOptimized) stochastically dominates the partial designs, raising the median from ~ 5.0 to ~ 6.7 bps/Hz; phase optimization accounts for most of the rate gain, while the learned power split adds a further, QoS-critical improvement. This is reflected in Fig. 5: the joint QoS-violation rate falls monotonically from 29.4% (no optimization) to 0.1% (proposed), with both the communication (q_c) and sensing (q_s) constraints satisfied almost surely.

Fig. 6 reports R_{sum} versus the RIS size $N \in \{8, 16, 32, 64\}$

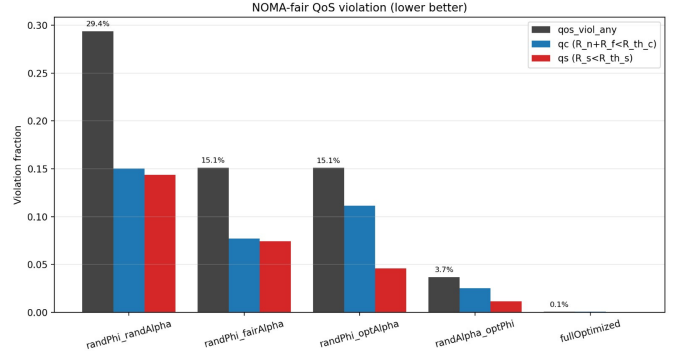
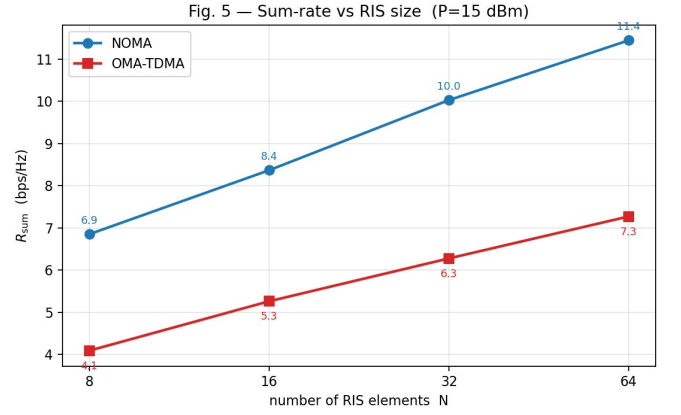


Fig. 5. QoS-violation fraction (lower is better). The joint violation rate drops from 29.4% to 0.1% along the ablation.


 Fig. 6. Weighted sum-rate versus RIS size N at $P = 15$ dBm. NOMA dominates OMA at every aperture; both scale with N .

at $P = 15$ dBm. The rate grows monotonically with N for both access schemes—from 6.9 to 11.4 bps/Hz for NOMA and from 4.1 to 7.3 bps/Hz for OMA—and NOMA retains an $\sim 65\%$ advantage throughout showing that the passive aperture and the access scheme contribute complementary gains.

For the OMA references, the resource variable is a time-allocation vector $\mathbf{t} = [t_n, t_f, t_T]$ rather than a NOMA power split. In the OMA-RIS branch, the same shared RIS phase is adapted in the inner loop, the network predicts $\mathbf{t}$ through a softmax layer, and the near/far slots use per-user MRT; the sensing slot uses the dominant sensing direction. In the OMA no-RIS branch, there is no phase variable, so the network predicts only the time fractions from the direct-link features while retaining per-user MRT and the sensing beam. Thus the OMA curves do not reuse the NOMA power-allocation policy.

Finally, Fig. 7 sweeps the transmit power. Panel (a) establishes the ordering NOMA-RIS $>$ NOMA $>$ OMA-RIS $>$ OMA across the whole range: NOMA's full-time superposition outperforms OMA's orthogonal slots, and the RIS adds a consistent margin on top of each access scheme (e.g. a $\sim 27\%$ RIS gain for NOMA at $P = 15$ dBm). Panel (b) shows the corresponding QoS-violation rate: NOMA-RIS reaches near-zero violation already at low power, whereas the no-RIS and OMA schemes need substantially more power to become feasible, quantifying the role of the RIS in jointly meeting the sensing and communication

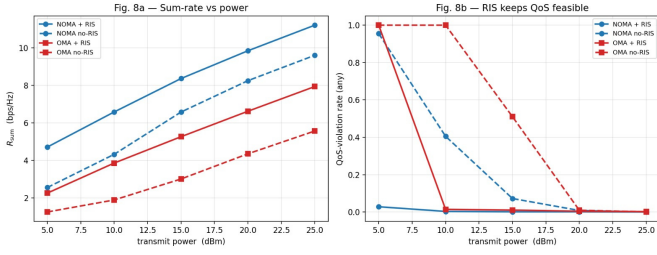


Fig. 7. (a) Weighted sum-rate and (b) joint QoS-violation rate versus transmit power for the four schemes.

QoS.

Finally, two practical points follow. On *runtime*, the online stage is a single forward pass of f_θ plus $K=5$ light inner steps—orders of magnitude below per-instance AO—so the design fits the short coherence times that motivate learning-based ISAC. On *constraint granularity*, replacing the sum-rate floor (11b) with per-user penalties ($\lambda_n, \lambda_f > 0$ in (14)) lets the operator dial the near/far fairness trade-off, shifting the learned split toward the weaker user at a modest cost in aggregate rate and with no change to the pipeline.

VI. CONCLUSION

A MISO RIS-assisted NOMA-ISAC downlink was investigated with joint power and phase design. The revised formulation combines an aggregate NOMA-cluster QoS constraint with individual near/far-user floors and a separate sensing floor, which preserves cluster efficiency while preventing a strong user from masking a weak one. The implemented solver uses a compact softmax power network and per-channel RIS-phase adaptation, followed by a first-order meta-update. The paper now reports the actual channel distances, fading/path-loss parameters, power/RIS settings, convergence limitations, and online prediction complexity. The original result figures and their surrounding discussion are retained and the figure files are provided in EPS form for submission.

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